

Requirements Specification

Team Project Module · National College of Ireland · Individual Project

Project Name: Smart Schedule

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1. Functional Requirements

Table 1: Functional Requirements, Priorities, and Acceptance Criteria for the Smart Schedule MVP

ID	User Story	Category	Priority	Acceptance Criteria
FR-001	As a staff member, I want to log in securely, so that only authorised users can access staff rota information.	User Management	MUST	Given a registered user with a valid email and password, when they submit the login form, then the system authenticates the user, creates a session, and redirects them to the correct role dashboard. Given invalid credentials, when the form is submitted, then access is denied and an error message is shown.
FR-002	As a manager, I want to create and manage staff accounts, so that employee details can be maintained in one system.	User Management	MUST	Given a manager is logged in, when they create a staff account with valid required details, then the new staff record is saved and appears in the staff list. Given an existing staff record, when the manager edits permitted details and saves the change, then the updated details are stored and shown in the staff list.
FR-003	As a user, I want to log out of the system, so that my account cannot be accessed after I finish using the application.	User Management	MUST	Given a logged-in user, when they select logout, then the active session is destroyed and the user is returned to the login page. Given the session has ended, when the user tries to open a protected page, then access is denied until they log in again.
FR-004	Weekly availability was part of the first design, but it is not part of the final Smart Schedule workflow.	Retired scope	RETIRED	Migration 014 removes the weekly availability table. Assignment checks now use active state, required role, approved leave, existing assignments, touching or overlapping shifts, and the five-shift / forty-hour weekly limits.
FR-005	As a manager, I want to create shifts for a selected week, so that rota assignments can be planned in advance.	Core Features	MUST	Given a manager is logged in, when they create a shift with a valid date, start time, end time, required role, and optional notes, then the shift is saved and displayed in the selected weekly rota view.
FR-006	As a manager, I want the system to block invalid assignments and warn when contract hours are exceeded, so that the weekly rota is checked before it is saved.	Core Features	MUST	Given a manager assigns or changes a shift, the backend blocks an inactive staff member, wrong role, approved leave, a duplicate assignment, an overlapping or touching shift, more than five shifts, or more than forty assigned hours in the Monday-to-Sunday week. Contract-hour excess is returned as a warning when the hard rules still allow the save.
FR-007	As a staff member, I want to request leave, so that managers can review and decide my time-off request.	Core Features	MUST	Given a staff member is logged in, when they submit a leave request with a valid start date, end date, and reason, then the request is stored with a pending status and is visible to managers for review.
FR-008	As a staff member, I want to view the full weekly rota, so that I can see my own shifts and who else is working.	UI/UX	MUST	Given a staff user is logged in, when they open Rota, the system returns the Monday-to-Sunday roster and open shifts without manager-only notes or edit controls.
FR-009	As a manager, I want to see assigned shifts and hours for the selected week, so that I can review workload before changing the rota.	Admin	SHOULD	Given the selected week contains assignments, the rota summary and staff rows show assigned shifts and calculated hours. The assignment service also reports contract-hour warnings and enforces the five-shift / forty-hour limits.
FR-010	As a manager, I want to export or print the rota, so that it can be shared outside the system if needed.	Integration	COULD	Given a rota exists for a selected week, when the manager selects export or print, then the system generates a readable rota output suitable for printing or sharing.

ID	User Story	Category	Priority	Acceptance Criteria
FR-011	As a manager, I want the system to calculate payroll, so that wages can be prepared automatically.	Admin	WON'T	Out of scope for this project iteration. Payroll is not implemented because wage rules, tax handling, and legal accuracy are outside this hospitality rota MVP.
FR-012	As a manager, I want staff to receive automatic SMS or push notifications, so that they are alerted when rota changes occur.	Integration	WON'T	Out of scope for this project iteration. NodyChat is in-app messaging only; SMS and push delivery are not claimed.
FR-013	As a staff member, I want to request a swap for my future assigned shift, so that another eligible staff member can take it after approval.	Core Features	MUST	The assignment owner can create a targeted or open request. An eligible staff member accepts it, a manager approves or rejects it, and the assignment changes only after a final eligibility check.
FR-014	As a manager, I want to preview the next rota from the current week, so that repeated shift patterns do not have to be entered one by one.	Core Features	SHOULD	The preview copies the current pattern, applies the live assignment rules, leaves unfilled shifts visible, and saves nothing until the manager approves the draft.
FR-015	As a signed-in user, I want to use NodyChat for workplace and direct conversations, so that rota questions stay inside the system.	Core Features	MUST	The shared WORKPLACE conversation and two-person DIRECT conversations require an active session. Only participants can load, send, receive, or mark direct messages read. A message contains 1 to 1000 trimmed characters and unread state is kept per user and conversation.
FR-016	As a user, I want to install the Smart Schedule web shell on a supported device, so that I can open it like an app.	UI/UX	COULD	The hosted frontend exposes a web manifest and service worker. Installation does not add offline rota editing and still uses the authenticated backend.
FR-017	As a user, I want to request a password reset without exposing whether my email exists, so that I can recover access safely.	User Management	MUST	The request returns the same public response for known and unknown addresses. A real reset token is hashed, expires, can only be used once, and the manager view does not expose the token or password.

Scope note: payroll, SMS/push notifications, POS integration, multi-branch support and full automatic scheduling remain outside this version. Weekly availability is retired rather than unfinished. Shift swaps, manager-reviewed rota population, NodyChat and the installable PWA shell are now part of the implemented baseline.

2. Non-Functional Requirements

Table 2: Non-Functional Requirements and Measurable Quality Targets for the Smart Schedule MVP

ID	Category	Requirement	Priority	Measurable Target
NFR-001	Performance	The application should load quickly on standard desktop and mobile devices.	SHOULD	Main pages should load in under 3 seconds on a normal broadband or 4G connection during manual testing.
NFR-002	Security	Passwords must be stored securely and must not be saved as plain text.	MUST	All stored user passwords are hashed with bcrypt before being written to the database, and no plaintext password is stored in any user record.
NFR-003	Security	Access to system features must depend on user role.	MUST	Staff users cannot access manager-only pages, routes, or actions, and unauthorised requests are rejected by the server.
NFR-004	Security / Data Integrity	The system must validate submitted data before saving it.	MUST	Required fields, date ranges, user identifiers, roles, and ownership checks are validated server-side before any insert or update is accepted.
NFR-005	Data Integrity	The system must store staff, leave, shifts, assignments, swaps, chat conversations, messages, and read states consistently.	MUST	Migrations 001 to 022 use foreign keys, unique rules, controlled statuses, and transaction-backed workflow updates to reject orphaned or invalid relationships.
NFR-006	Usability	The main rota and staff actions should stay usable for non-technical hospitality staff.	MUST	Rota, time off, swaps and NodyChat use the same signed-in shell and must remain keyboard reachable, readable at supported browser sizes, and usable at 200% zoom during manual checks.
NFR-007	Compatibility	The system should work on current desktop and mobile browsers.	SHOULD	The responsive layout must remain readable at the recorded desktop, tablet and phone widths without horizontal loss of forms, navigation, dialogs, or the weekly rota controls.
NFR-008	Security	Conversation access must be checked on the server for HTTP and WebSocket actions.	MUST	Direct messages are returned and broadcast only to participant user IDs. Cross-origin WebSocket upgrades, inactive sessions and read updates for non-participant messages are rejected.

3. Reference

Table 3: Requirements Reference Guide

Area	Term / Category	Meaning / Use
Priority	MUST	Essential for the MVP. These requirements must be implemented for the system to meet its core purpose.
Priority	SHOULD	Important but not vital. These requirements add value but can be delayed if time becomes limited.
Priority	COULD	Useful extra features that are not required for the MVP.
Priority	WON'T	Out of scope for this iteration and not planned for implementation in the current MVP.
Functional Category	User Management	Requirements related to login, logout, user roles, permissions, and staff account management.
Functional Category	Core Features	Main application features such as rota creation, availability, leave requests, conflict checks, and shift assignment logic.
Functional Category	Data	Requirements related to storing, retrieving, updating, and protecting application data.
Functional Category	Integration	Requirements involving external outputs or services such as export, print, notifications, or APIs.
Functional Category	UI/UX	Requirements related to visibility, navigation, layout, and user interaction.
Functional Category	Admin	Requirements related to manager reporting, summaries, and control functions.
Non-Functional Category	Performance	Requirements related to speed, loading time, and responsiveness.
Non-Functional Category	Security	Requirements related to authentication, password protection, access control, and secure handling of user actions.
Non-Functional Category	Data Integrity	Requirements related to keeping stored system data valid, consistent, and protected from invalid relationships.
Non-Functional Category	Usability	Requirements related to clarity, ease of use, and user efficiency.
Non-Functional Category	Compatibility	Requirements related to working correctly across supported devices and browsers.
Acceptance Criteria	Given / When / Then	A testable structure where a context is defined, an action occurs, and an observable result must happen.
Priority	RETIRED	A requirement from an earlier design that was deliberately removed from the final baseline. It is not counted as an unfinished MUST.